Database Development and Design (SOE201)

Lab 5: Tutorial on Query Optimisation

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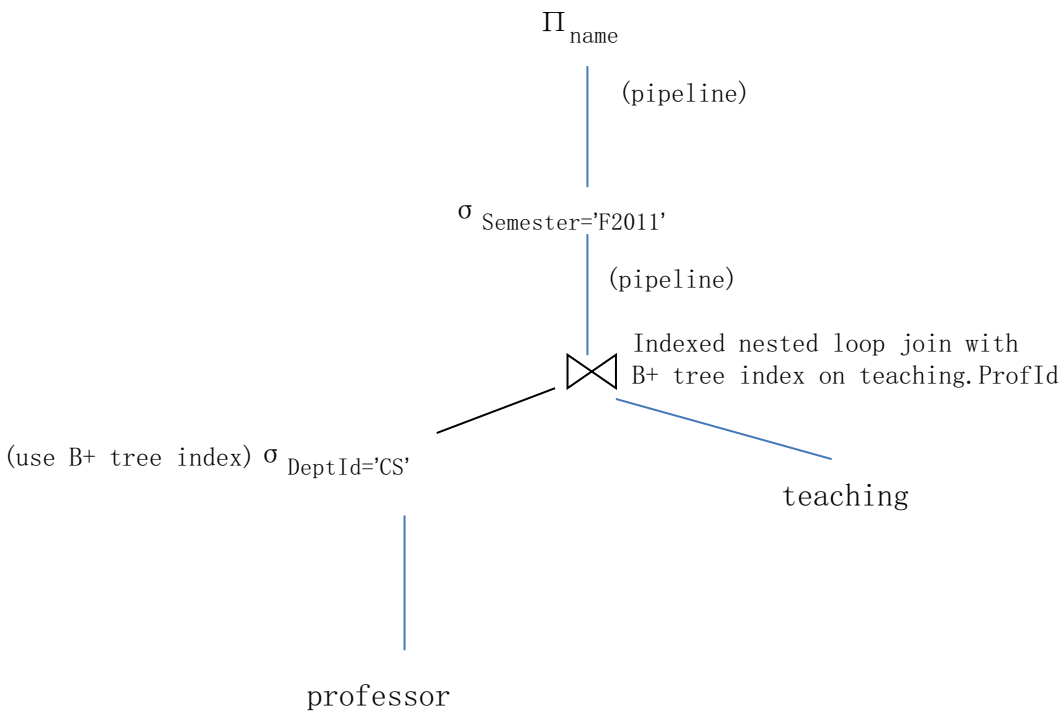
AIAT

Q1

- Given two relations *professor*(*ProfId, Name, DeptId*) and *teaching*(*ProfId, CourseCode, Semester*) with the following catalog information:
  - $n_{professor}$ = 1,000
  - $b_{professor}$ =200
  - index: a clustered 2-level B+ -tree on *DeptId* for relation *professor*.
  - V(*DeptId*, professor) = 50
  - $n_{teaching}$ = 10,000
  - $b_{teaching}$ =1,000
  - index: a B+-tree index on ProfId for relation teaching.
- Consider the query evaluation tree below. Assume that the average cost (the number of block transfers) of using the B+-tree index to fetch all matching tuples from relation *teaching* for one tuple in professor is *X*. Furthermore, no block transfers are needed for an operation when pipelining is used. Answer the following questions.
  - (a). What is the estimated size of evaluating $\delta_{DeptID='CS'}$ Professor?
  - (b). What is the estimated number of block transfers for $\delta_{DeptID='CS'}$ Professor using the B+-tree?
  - (c). Based on the answers you have for $\delta_{DeptID='CS'}$ Professor, what is the estimated number of block transfers for evaluating the *join* operation?
  - (d). What is the total number of block transfers for the whole evaluation plan?

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Q1 cont'd


Π name
(pipeline)
σ Semester='F2011'
(pipeline)
Indexed nested loop join with
B+ tree index on teaching.ProfId
(use B+ tree index) σ DeptId='CS'
teaching
professor

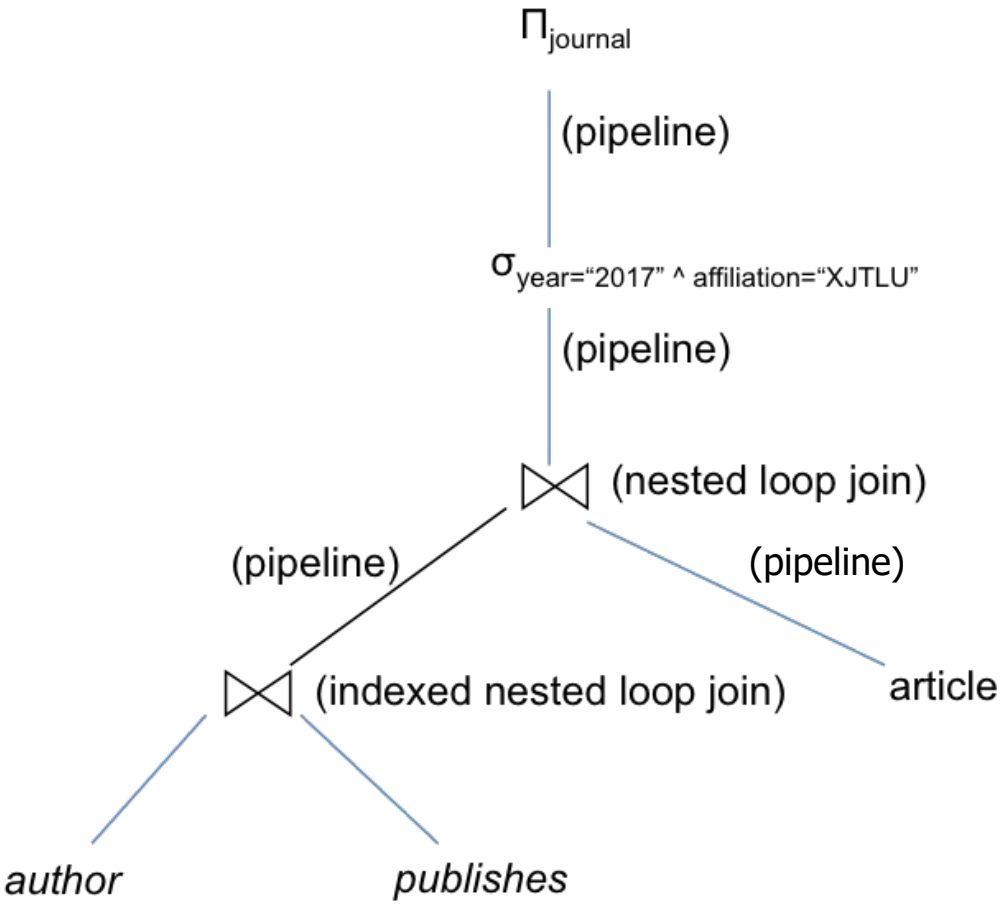
Q2

- Consider the following three relations, evaluation plan and their catalog information. "*authorId*" and "*articleID*" are the candidate keys for relations *author* and *article*, respectively. The two attributes in *publishes* are the foreign keys referencing *author* and *article*, respectively.
  - *author (authorID, name, affiliation, email)*
  - *publishes (articleID, authorID)*
  - *article (articleID, title, journal, year, publisher)*
- Catalogue information is as follows
  - Number of records in *author*, n_{author} = 5,000;
  - Number of blocks in *author*, b_{author} = 500;
  - Number of records in *publishes*, $n_{publishes}$ = 100,000;
  - Number of blocks in *publishes*, $b_{publishes}$ = 100;
  - Number of records in *article*, $n_{article}$ = 20,000;
  - Number of blocks in *article*, $b_{article}$ = 2,000;
  - A primary B^+-tree index of height 3 on the *authorID* attribute of *publishes* relation;
  - Number of distinct values for the attribute *affiliation* in the *author* relation, *V(affiliation, author) = 500;*
  - Number of distinct values for attribute *year* in relation *article*, *V(year, article) = 10;*

Q2 cont'd

- Consider the query evaluation tree below and answer the following questions.
  - 1) One of the heuristic rules for query optimisation is to perform selection as early as possible. In the above query tree, the selection predicate *affiliation*="XJTLU" can be performed on relation *author* first and *year*="2017" can be performed on relation *article* first. Draw the equivalent evaluation tree based on this heuristic rule and the equivalence rules for algebra expressions.
  - 2) Based on your answer from Question 1), what would be the estimated size of the indexed nested loop join?
  - 3) Based on your answer from Question 1), assume that linear scan is used to evaluate all the selection operations in the plan. What is the total number of block transfers for the optimised evaluation plan? Note that no intermediate relations need to be saved as the result of using pipelining.

Q2 cont'd


Π journal
(pipeline)
σ year="2017" ^ affiliation="XJTLU"
(pipeline)
(nested loop join)
(pipeline)
(pipeline)
(indexed nested loop join)
article
author
publishes

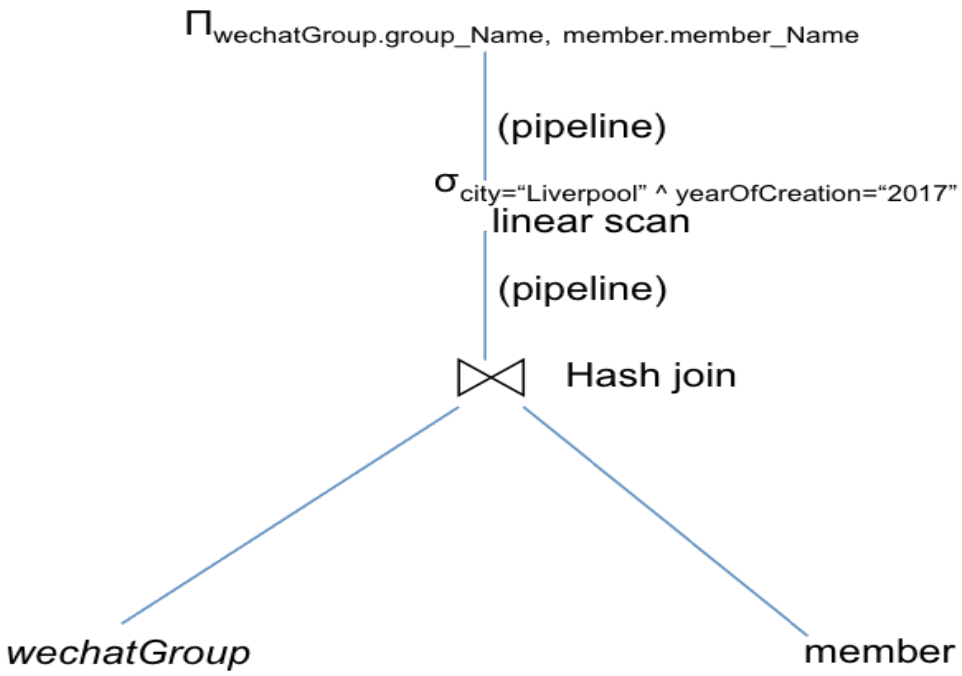
Q3

- Consider the following two relations and their catalog information.
  - *wechatGroup*(*gourp_ID*, *group_Name*, *size*, *yearOfCreation*, *owner*)
  - *member*(*member_ID*, *member_Name*, *city*, *group_ID*)
  - *group_ID* and *member_ID* are the keys for the two relations, respectively.
  - Number of tuples in relation *wechatGroup*, n_r = 1,000
  - Number of blocks in *wechatGroup*, b_r =200
  - Number of distinct values on attribute *yearOfCreation* in *wechatGroup*, V(*yearOfCreation*, *wechatGroup*) = 100
  - Number of tuples in relation *member*, n_s = 10,000
  - Number of blocks in relation *member*, b_s =1,000
  - Number of distinct values on attribute *city* in *member*, V(*city*, *member*) = 100

Q3 cont'd

- Consider the query evaluation tree below and answer the following questions.
  - (1) Write the relational algebra expression for the given evaluation plan.
  - (2) One of the heuristic rules for query optimisation is to perform selection as early as possible. Write the equivalent algebra expression for the answer from Question Q3.(1).
  - (3) Suppose that all selections are evaluated by using linear scan and pipelining is used for projection. Draw an annotated evaluation tree based on your answer for Question Q3.(2).
  - (4) Assume that for each tuple in $\sigma_{yearOfCreation="2017"}(wechatGroup)$, the average cost of performing the hash join is 3 block transfers (including all the block transfer for relation partition and the build/probe procedure). What would be the total number of block transfers for the whole evaluation plan in Question Q3.(3)?

Q3 cont'd


Π wechatGroup.group_Name, member.member_Name
(pipeline)
σ city="Liverpool" ^ yearOfCreation="2017"
linear scan
(pipeline)
Hash join
wechatGroup
member

Q4

- The database of an online movie website has the following three relations.
  - *performer (pid, name, sex, agent, country, email)*
  - *performsIn (pid, mid)*
  - *movie (mid, title, year, language, producer)*
- The attributes "*pid*" and "*mid*" are the keys for relations *performer* and *movie*, respectively. The two attributes in *performsIn* are the foreign keys referencing *performer* and *movie*, respectively. An agent for a performer is someone who signs a contract with and represents the performer. The catalogue information is given below.
  - In relation *performer*, number of records $n_{performer}$ = 3,800; number of blocks $b_{performer}$ = 520;
  - In relation *performsIn*, number of records $n_{performsIn}$ = 70,050; number of blocks $b_{performsIn}$ = 86;
  - In relation *movie*, number of records, n_{movie} = 31,000; number of blocks b_{movie} = 2,700;
  - Index: a primary B^+-tree index of height 7 on the *pid* attribute of the *performsIn* relation;
  - Number of distinct values for the attribute *agent* in the *performer* relation *V(performer, agent) = 500;*

Q4 cont'd

- Consider the query evaluation plan below and answer the following question. Assume that linear scan is used to evaluate all the selection operations, and no intermediate relations need to be stored as the result of using pipelining. What would be the total number of block transfers? Justify your answer.

